The Chain Atom $W = u^5v + v^6$, the Missing Leading Term v^{11} in its Berglund–Hübsch Transpose $W^T = u^5 + uv^6$, and Affine Milnor Fibre vs. Projective Fourfold Cohomology

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Abstract

This paper synthesizes the three preprints in BenFrohman/SingularityLab (PREPRINT_FINDINGS.md, PREPRINT_V11.md, and PREPRINT_AFFINE_PROJECTIVE.md) investigating the invertible chain polynomial $W(u,v) = u^5v + v^6$, its Berglund–Hübsch transpose $W^T(u,v) = u^5 + uv^6$, and the three-block Thom–Sebastiani sextic fourfold $F = \sum_{i=0}^{2}(x_i^5 x_{i+3} + x_{i+3}^6) \subset \mathbb{P}^5$. We prove and Lean-certify (lean/MilnorCount.lean, LedgerIntegers.lean, LedgerCert.lean) the invariant separation

$$\mu(W) = 25, \qquad \mu(W^T) = 26, \qquad |\det A| = |\mathrm{Aut}(W)| = |\mathrm{Aut}(W^T)| = 30,$$

with inner modalities $m_0(W) = 6$ and $m_0(W^T) = 5$, explain the Gröbner basis mechanism by which the pure power $v^{11} \in (\partial W^T)$ is eliminated in one step ($v^{11} = v^5\partial_u W^T - \frac{1}{5}u^4\partial_v W^T + \dots$), and establish the container separation between the affine Milnor fibre ($\mu(F) = 25^3 = 15625$) and the projective primitive cohomology ($h^{2,2}_{\mathrm{prim}}(X) = \dim R_{12}(F) = 1751$).

1 Main Invariant Theorems

Theorem 1 (Chain Atom vs. Berglund–Hübsch Transpose)**.** *For $W = u^5v + v^6$ with exponent matrix $A = \left(\begin{smallmatrix} 5 & 1 \\ 0 & 6 \end{smallmatrix}\right)$ and transpose $W^T = u^5 + uv^6$ with $A^T = \left(\begin{smallmatrix} 5 & 0 \\ 1 & 6 \end{smallmatrix}\right)$:*

1. *$|\det A| = |\det A^T| = |\mathrm{Aut}(W)| = |\mathrm{Aut}(W^T)| = 30$, and central charge $\hat{c}(W) = \hat{c}(W^T) = 4/3$.*
2. *$\mu(W) = (6-1)^2 = 25$, whereas W^T is quasihomogeneous of weights $(3,2)$ and degree 15 with $\mu(W^T) = (15/3 - 1)(15/2 - 1) = 26$.*
3. *In $\mathbb{C}[u,v]/(\partial W^T)$ where $\partial_u W^T = 5u^4 + v^6$ and $\partial_v W^T = 6uv^5$, the relation $v^6 \equiv -5u^4$ implies $v^{12} \equiv 25u^8$ and $uv^5 = 0 \implies v^{11} = v^5(-5u^4) = -5u^3(uv^5) \equiv 0$, cutting the monomial box at v^{11} to yield dimension 26.*

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